

# Module 6 Assignment

## Description

There are two distinct parts to this assignment. In the first part you will complete some exercises designed to get you accustomed to using bindings. In the second part you will build a very small application that uses a UI model to control its contents.

## First steps

Make sure you've exited IntelliJ completely. Using the command line, in your solutions directory,

1. switch to the main branch and merge in your module5 branch.
2. create (and switch to) a branched name module6.

Start IntelliJ and create a new IntelliJ project using the JavaFX template. Name the project module6 and place it inside your solutions directory. **IMPORTANT:** disable the "create git repository" checkbox since you've already created a repository.

The JavaFX templates creates several files (two .java files and an fxml file). As this project proceeds, please rename these files to something more appropriate for this assignment. For example, in my solution I renamed HelloApplication.java to FlightScheduleApplication.java.

Be sure to commit and push regularly as you work through this assignment.

## Part 1: Practice with properties and bindings

Create a package called edu.au.cpsc.part1. Download and unzip [Part1Files.zip](#) [Download Part1Files.zip](#). Copy Part1Controller.java and Part1Application.java into that package. Create a resources directory with path edu/au/cpsc/part1 (IntelliJ will show it with dots instead of slashes but create the directory with the slashes). Copy part1-view.fxml and img.png into that resources directory.

Do not make any changes to these files except Part1Controller. In Part1Controller place all of your code in the initialize() method. Do not otherwise modify this class. **Do not use event handlers of any form, only property bindings.**

- Create a property binding so that whatever is typed in messageTextField also shows in echoTextField.
- Create a bi-directional property binding so that whatever is typed in firstBidirectionalTextField shows in secondBidirectionalTextField and vice versa.
- Create a binding so that the opacity of the secretOverlapImageView is set by the value of the secretSlider.

- Create a binding so that the selectMeLabel will display true/false depending on the state of selectMeCheckBox. Hints: CheckBox has a selectedProperty that is a Boolean property. Labels have a textProperty but this must be some kind of string-like property (not a String!). Most properties in JavaFX have an asString() method that returns a string-like property of some type. The difference between asString() and toString() is that the former returns a property while the latter returns a String (which cannot be used for binding).
- Create a binding so that numberOfCharactersLabel displays the number of letters in the tweetTextField. You will need length() and asString().
- Create a binding so that validityLabel displays "Valid" if there are 10 characters or fewer in tweetTextField and "Invalid" otherwise. Hint: Use Bindings.when().

## Part 2: UI model

Copy your Java and FXML files from your module 4 project into this IntelliJ project. If your module 4 project isn't working, you'll need to create a simple editor application that you can use to implement the features described here (something like the module 2 project would be OK). Create a UI model and use it in your detail editor and add the following features:

- Enable/disable buttons based on state of the model object (new/modified).
- Place validation logic in your UI model and use it to highlight invalid input values (one simple way to do this is to have a boolean property for each input that indicates if the input is valid or not and use that property to change some visual feature of the UI).

## Submitting

Commit and push one last time and verify that your files look OK on GitHub (make sure you switch to the correct branch when looking at it on GitHub). Next submit this assignment on Canvas by pasting the URL of your GitHub repository into the supplied input. This will be the same URL that you submitted in Module 1. Note that Canvas will try to display a preview of it but that preview may show an error since the repository is private. Don't worry, as long as your URL is correct, everything is OK.